A2 UX Design – Car Rental App

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This document holds the UX material the brief asks for, updated for the final Fake Uber design (v8, 14 screens): two user stories, two use cases, two low-fi layouts with their widgets, and the reasons for choosing Layout A. It also fixes the five cars the app will show and the final colour values.

1. User stories

The two stories cover the app's two core needs: **choosing a suitable car** and **renting it without overspending**. Each acceptance criterion can be checked by a person or turned into an Espresso test in phase 5.

US1: Browse, find and compare cars

> *As a customer looking for a short rental, I want to see each car's price, rating, year and kilometres under its photo, and to search, sort and save cars, so that I can pick the right one before I start a booking.*

Acceptance criteria:

1. Home shows one car at a time with brand, model, year, rating (stars + number), kilometres and price per day under the photo, plus a "1 / 5" position counter.
2. Next moves to the next available car; rented cars are skipped.
3. Typing in the search field keeps only cars whose brand or model matches, case-insensitively. No match shows "No cars match '…'" with a Clear search button.
4. The Sort menu offers Rating (high to low), Year (newest first) and Cost (lowest first), ticks the active option, and reorders a copy of the list, never the base data.
5. Tapping the heart or long-pressing the photo adds the car to Favourites; tapping again removes it.

US2: Rent within my credit, with my documents

> *As a customer with 500 credits, I want to see the total cost and my balance after rental before I confirm, and to add my ID and licence photos in the same place, so that I never book a car I cannot afford or am not allowed to drive.*

Acceptance criteria:

1. The credit balance is visible in the header of Home and Rent.
2. On Rent, the total (price × days), the balance after rental and the 400-credit limit bar update as soon as the 1–7 day slider moves.
3. Save is refused, with the reason shown in place, if the total is over 400, over the balance, or either document photo is missing. The user stays on Rent.
4. A valid Save returns to Home, deducts the total, removes the car from Available now, shows it under My rentals and shows "… booked" in a Snackbar.
5. Back on Rent saves nothing and shows "Rental not saved". Cancel booking in My rentals refunds the credits, returns the car to the list and shows "Booking cancelled · … refunded".

2. Use cases

A user story says **what** the user wants; a use case walks through **each step** between user and app, including the error branches. UC1 implements US2 and UC2 implements US1, and together they make a good outline for the demo.

UC1: Rent a car

Field	Description
Actor	Customer
Goal	Rent a car for 1–7 days without breaking the credit rules
Precondition	Home is open and at least one car is available
Success	Credits deducted, car listed under My rentals and removed from Available now
Cancel	Nothing changes; credits stay the same

Main flow

1. The customer presses **Next** until the wanted car is shown, then presses **Rent**.
2. The app opens the Rent screen and passes the car in an Intent as a Parcelable.
3. The app shows the photo, details, the slider at 1 day, the Credits card and two empty document tiles.
4. The customer moves the slider; the total, balance after rental and limit bar update at once.

5. The customer taps **Driver ID Photo** and **Driver Licence Photo** and picks a photo for each in the Photo Picker. Each tile shows a tick.
6. The customer presses **Save · N credits**.
7. The app checks: total ⩽ 400, total ⩽ balance, both photos present.
8. The app returns to Home, deducts the credits, moves the car to My rentals and shows the Snackbar "Toyota Vios booked · 3 days · 240 credits".

Alternative flows

- **7a. Total over 400:** the limit bar turns red, the message "This rental costs 440 credits. One rental can't go over 400…" appears in the Credits card, and the Snackbar "Can't save yet" appears. The customer picks fewer days and returns to step 6.
- **7b. Total over balance:** the message "Not enough credits. You have 170, this rental needs 240…" appears. The customer picks fewer days or presses Back.
- **7c. A photo is missing:** the empty tile turns red with "Required". The customer adds the photo and returns to step 6.
- **3–7. Back pressed:** the app returns to Home with no change and shows "Rental not saved".
- **After 8. Cancel booking:** from My rentals, the customer cancels; the app refunds the credits, returns the car to the list and shows "Booking cancelled · 240 credits refunded".

UC2: Find, sort and save a favourite

Field	Description
Actor	Customer
Goal	Find a car by name or criterion and keep it for later
Precondition	Home is open
Success	The car appears in the Favourites row with a filled heart

Main flow

1. The customer types part of a name or model in the search field, for example "vios".
2. The app filters as the customer types and the counter changes to "1 / 1".
3. The customer clears the search, opens **Sort** and picks "Cost · lowest first".
4. The app reorders a copy of the list and ticks the active option; the base list is unchanged.

5. The customer taps the heart on the photo.
6. The app fills the heart and adds a chip (thumbnail, name, price) to Favourites; the count reads "1 saved".

Alternative flows

- **2a. No match:** the app shows "No cars match 'xyz'", the counter "0 / 0" and a Clear search button.
- **5a. Long-press on the photo:** same result as step 6, offered as a shortcut.
- **5b. Already a favourite:** tapping the heart removes it from Favourites.

3. The five cars

The fleet copies what Vietnamese self-drive rental services actually list, one car per popular segment, priced at **1 credit = 10,000 VND**. Prices come from published 2026 rental price lists; ratings and kilometres are illustrative, because no source publishes them per car.

What the market shows. BonbonCar's April 2026 price list puts a Toyota Vios 2024 at 800,000 VND a day, a Mazda3 at 950,000–1,000,000, a Mitsubishi Xpander 2025 at 1,100,000 and a VinFast VF 5 Plus 2025 at 900,000 (BonbonCar, 2026a). Its Ho Chi Minh City VinFast list gives the VF 5 Plus at 400,000–600,000 (BonbonCar, 2026b). Tiền Phong reports normal-day prices of 950,000–1,100,000 for a Toyota Fortuner, 650,000–750,000 for an Xpander and 500,000–650,000 for a VF 5, with SUVs and 7-seaters the most in demand before holidays (Tiền Phong, 2026). Dân Trí's Tết survey lists the same models (VF 5, Vios, Accent, City, Mazda3, Xpander, Fortuner) across segments A to D and MPV (Dân Trí, 2026).

Brand	Model	Year	Type	Seats	Fuel	Km (est.)	Rating	Price	Market price (VND/day)
VinFast	VF 5 Plus	2025	Electric SUV	5	Electric	18,500	4.6	60 cr	400k–900k
Toyota	Vios 1.5G	2024	Sedan	5	Petrol	41,000	4.8	80 cr	800k
Mazda	Mazda3 1.5	2022	Sedan	5	Petrol	74,000	4.7	90 cr	950k–1.0M
Mitsubishi	Xpander AT	2023	MPV	7	Petrol	58,000	4.5	100 cr	650k–1.1M
Toyota	Fortuner 2.4AT	2021	SUV	7	Diesel	92,000	4.9	110 cr	950k–1.1M

Three design choices sit behind these numbers:

- **Kilometres** assume about 17,000 km a year, inside the 9,000–24,000 km a normally used car covers (Ngôi Sao, 2025), slightly high because rental cars are driven more.
- **The three sort orders all differ**, so the demo shows that sorting works: rating puts the Fortuner first, year puts the VF 5 first, and cost puts the VF 5 first and the Fortuner last.
- **The limits can be shown with real cars**: 4 days of Fortuner costs 440 (over the 400 limit), 7 days of VF 5 costs 420, and after a 3-day Fortuner rental (330) only 170 credits remain, too few for 3 days of Vios (240).

Photos

All five photos come from Wikimedia Commons under Creative Commons licences, which allow free use with credit. Credit each one in the app's About text or the report. The originals are high-resolution; resize to about 1280 px wide before adding them to `res/drawable`.

Car	Commons file	Author	Licence
VinFast VF 5 Plus	2025 VinFast VF 5 in Vinfast Blue, front left	Ethan Llamas	CC BY-SA 4.0
Toyota Vios	Toyota Vios XP150 Facelift Shishi 01 2022-09-20	JamesYoung8167	CC BY-SA 4.0
Mazda3	MAZDA3 SEDAN (BP) China	Dinkun Chen	CC BY-SA 4.0
Mitsubishi Xpander	2022 Mitsubishi Xpander Ultimate 1.5 NC2W	オーバードライブ83	CC BY-SA 4.0
Toyota Fortuner	2021 Toyota Fortuner 2.4 TRD Sportivo (Indonesia) front view	Andra Febrian	CC BY-SA 4.0



1. VinFast VF 5 Plus



2. Toyota Vios



3. Mazda3 Sedan



4. Mitsubishi Xpander



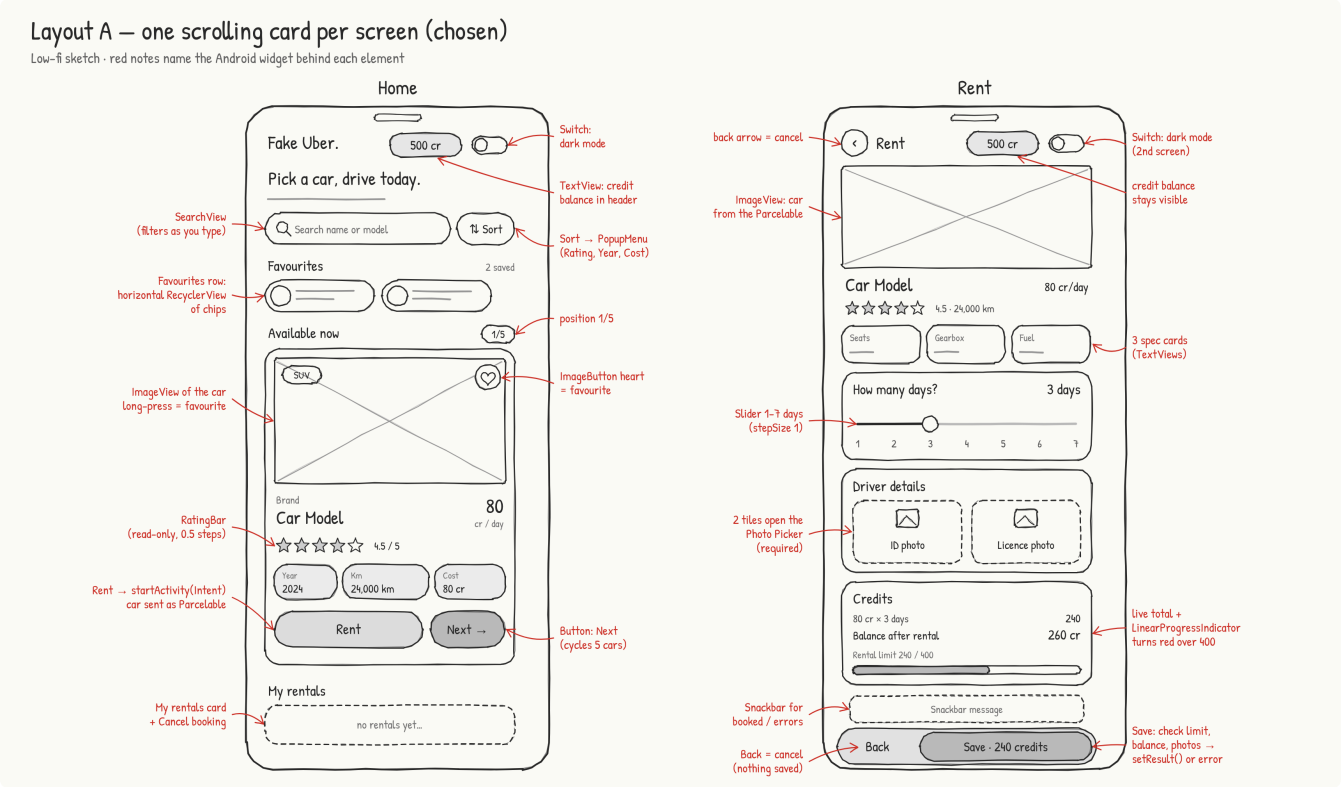
5. Toyota Fortuner

The photos are of models sold in Vietnam but were taken abroad, so number plates differ; the plates are small enough not to matter at card size.

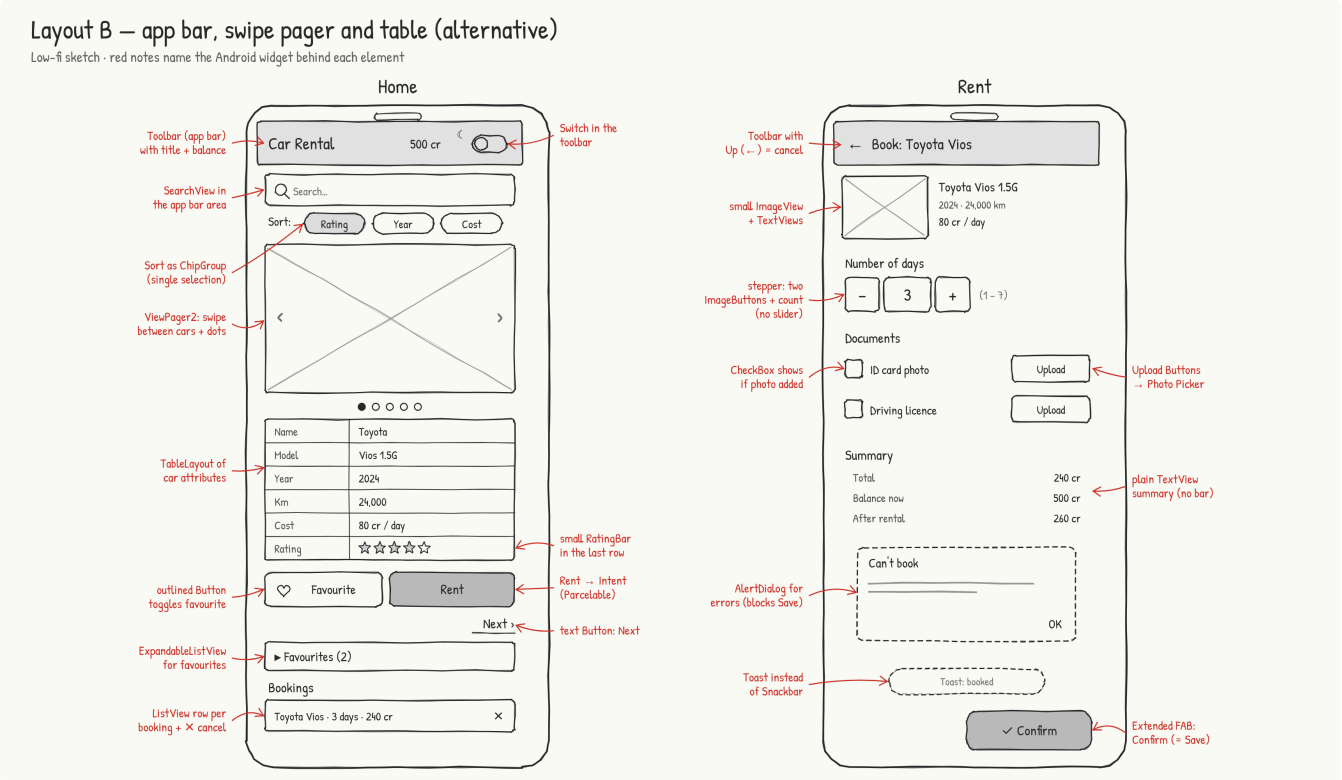
4. Two low-fi layouts

The brief asks for sketches of two possible layouts that use different UI elements, with the required widgets noted. Both sketches below are deliberately rough: grey boxes, crossed image placeholders and red notes naming the Android widget behind each element. Each shows the two screens the app needs, Home and Rent.

Layout A is the final Fake Uber design reduced to a wireframe. Home stacks titled blocks in one scrolling column: header with credit balance and dark-mode Switch, an always-visible search field with a Sort button, a Favourites row, one car card with Rent and Next buttons, and My rentals. Rent repeats the car, then three cards in decision order: days (Slider), driver documents (two photo tiles) and credits (live total and limit bar), with a fixed Back/Save bar.



Layout B answers the same brief with a more "standard Android" structure. A Toolbar holds the title, balance and Switch; cars are swiped in a ViewPager2 with dots and a text Next button; sorting is a ChipGroup; details sit in a TableLayout; favourites collapse into an ExpandableListView and bookings into a ListView. Rent uses a stepper (− / +) instead of a slider, Upload buttons with CheckBoxes for documents, a plain text summary, an AlertDialog for errors, a Toast for confirmation and an Extended FAB to confirm.



Element	Layout A	Layout B
Moving between cars	Next button + "1 / 5" counter	ViewPager2 swipe + dots + text Next
Sort	Button opening a PopupMenu	ChipGroup, always visible
Car details	Stat chips + RatingBar under the photo	TableLayout rows + small RatingBar
Favourites	Horizontal row of chips	ExpandableListView
Bookings	My rentals card with Cancel booking	ListView row with ×
Number of days	Slider 1–7	Stepper: two ImageButtons + count
Driver documents	Two large photo tiles	Upload Buttons + CheckBoxes
Cost	Credits card with live limit bar	Plain TextView summary
Errors	Inline messages + Snackbar	AlertDialog
Confirmation	Snackbar	Toast
Save	Pill button in a fixed bottom bar	Extended FAB

Both layouts keep every requirement: one car at a time with Next, balance at the top, Rent opening a new Activity with a Parcelable, a dark-mode Switch on both screens, and at least two non-TextView widgets (A: RatingBar, Slider, Switch; B: RatingBar, stepper, Switch).

5. Comparison and choice

Layout A is chosen. It wins on six of the eight criteria below; Layout B is better only at sorting in one tap and at fitting more details on screen, and both advantages are small for a five-car app.

Criterion	Layout A	Layout B	Better	Evidence
Discoverability of moving between cars	Visible Next button and a labelled counter	Swipe is hidden; dots are weak cues	A	Budiu, 2018; Kweon et al., 2026
Taps to sort	2 (open menu, choose)	1 (chip)	B	Laws of UX, Hick's Law

Criterion	Layout A	Layout B	Better	Evidence
Preventing over-limit rentals	Live total and limit bar warn before Save	Summary text only; error appears after Save	A	Nielsen heuristic 5, error prevention
Quality of error messages	In place, specific numbers, Save still explains	Modal dialog blocks the screen	A	Neusesser and Sunwall, 2023
Feedback that can be built as drawn	Bottom Snackbar, read by TalkBack	Toast, cannot be positioned on Android 11+	A	Material Components, Snackbar; Künneth, 2020
Choosing days	One drag across a visible 1–7 range	Up to 6 taps on small −/+ targets	A	Parhi et al., 2006
Information density	Card shows the key facts; rest on Rent	Table shows every attribute at once	B	Recognition over recall (Budiu, 2024)
Espresso testing	Click Next, Rent, Save; check text	Swipes and dialogs need extra handling	A	The brief advises testing text and buttons

Justification for the report

> *Layout A was chosen because every action is a visible control within thumb reach, and because it prevents mistakes instead of reporting them. Layout B's swipe navigation is a hidden gesture, its stepper needs several small taps to reach a week, and its errors arrive in a modal dialog only after Save. Layout A's Credits card shows the total, the balance after rental and the 400-credit limit while the user is still choosing, and its errors appear in place with the exact numbers. B's chip-based sorting is one tap faster and its table shows more at once, but with only five cars and three sort options the saving is small. Layout A also keeps feedback in a bottom Snackbar, which can be built exactly as drawn, whereas a positioned Toast cannot on Android 11 and later.*

Rewrite this paragraph in your own words for the report. If you test the app with a few classmates, add what you observe; real observations are the strongest evidence.

6. Final colours and style plan

The accessibility fixes from the UI/UX report are adopted, so the app is built with the corrected values from the start. Only five tokens change; every other colour stays as designed.

Token	Was	Final	Why
`orange` (accent, Rent/Save fill, slider, heart)	#F97316	#EA580C	Orange text and icons reach 3.1–3.4:1 instead of 2.5–2.7:1; dark text on the button still 5.04:1
`text_secondary` (light)	#716C65	#625D57	Hints on tiles and chips reach 5.25–5.44:1 instead of 4.19–4.35:1
`green` ("RENTED" label)	#2F9E5E	#1F7A45	5.17:1 instead of 3.29:1
`error_text` (light, inside the pink box)	#E11D2E	#B91C1C	5.32:1 instead of 3.91:1
`error_text` (dark theme)	#E11D2E	#F87171	5.98:1 instead of 3.48:1

Colours live in `values/colors.xml` and `values-night/colors.xml` and are reached through theme attributes, so the dark-mode Switch changes both screens at once. The app theme `Theme.FakeUber` extends `Theme.Material3.DayNight.NoActionBar`, and every touch target gets a 48dp minimum.

The brief asks for at least two styles, each used in at least two places. The plan has six, all used on both screens:

Style	Defines	Used on Home	Used on Rent
`TextAppearance.FakeUber.Title`	22sp extra-bold Inter, primary text colour	Car model on the card	Car model under the photo
`TextAppearance.FakeUber.Section`	16sp bold, primary text colour	Favourites, Available now, My rentals	How many days?, Driver details, Credits

Style	Defines	Used on Home	Used on Rent
`TextAppearance.FakeUber.Secondary`	13sp regular, `text_secondary`	Brand, stat labels, hints	Meta line, helper text, credit rows
`Widget.FakeUber.CreditPill`	Dark pill, wallet icon, balance text	Header	Header
`Widget.FakeUber.Button.Primary`	Orange fill, dark label, 48dp min height, pill shape	Rent	Save
`Widget.FakeUber.Card`	Surface colour, 24dp corners, 1dp stroke	Car card, My rentals card	Days, Driver details and Credits cards

Example, ready to paste into `values/styles.xml`:

```
<style name="Widget.FakeUber.Button.Primary"
parent="Widget.Material3.Button">
    <item name="backgroundTint">@color/orange</item>
    <item name="android:textColor">@color/on_orange</item>
    <item name="android:minHeight">48dp</item>
    <item
name="shapeAppearance">@style/ShapeAppearance.Material3.Corner.Full</item>
</style>
```

Sources

All sources are free to read and were accessed on 7 November 2026. Design evidence is discussed in full in the companion report, A2 UI/UX Research Report – Fake Uber Final Design.

1. BonbonCar (2026a) Bảng giá dịch vụ thuê xe tự lái 4 chỗ, 7 chỗ, updated 20 April 2026.
2. BonbonCar (2026b) Bảng giá thuê xe tự lái VinFast tại TP.HCM.
3. Dân Trí (2026) Khảo giá thuê ô tô tự lái Tết 2026 từng phân khúc, 11 February 2026.

4. Tiền Phong (2026) Thuê ô tô tự lái dịp lễ: xe phân khúc nào đắt khách nhất?, 28 August 2026.
5. Ngôi Sao (2025) Xe ô tô đi được bao nhiêu ODO thì được xem là quá cũ?, 2 March 2025.
6. Wikimedia Commons, the five photo pages listed in section 3.
7. Budiu, R. (2018) Carousels on Mobile Devices. Nielsen Norman Group.
8. Budiu, R. (2024) Memory Recognition and Recall in User Interfaces. Nielsen Norman Group.
9. Kweon, M. et al. (2026) GhostUI: Unveiling Hidden Interactions in Mobile UI. CHI '26.
10. Künneth, T. (2020) Care for a toast in the snackbar? DEV Community.
11. Laws of UX (n.d.) Hick's Law.
12. Material Components for Android (n.d.) Snackbar documentation.
13. Neusesser, T. and Sunwall, E. (2023) Error-Message Guidelines. Nielsen Norman Group.
14. Parhi, P., Karlson, A. K. and Bederson, B. B. (2006) Target Size Study for One-Handed Thumb Use on Small Touchscreen Devices. MobileHCI '06.